

Context Is the New Weight: Trading the Attention Sink for a Constant Working Memory via Windowed Finetuning

Anonymous submission

Abstract

Causal language models develop an *attention sink*: heads route a large fraction of their attention onto the first one or two tokens, regardless of content. We argue this sink is not an accident but a direct consequence of the triangular causal mask — early positions are visible to every later query, so they accumulate attention — and that it is precisely what makes a pretrained model fragile to a bounded context: naively restricting attention to a sliding window deletes position 0 and the model collapses. We instead finetune the attention itself under a sliding-window mask, so the model operates with a *constant working memory* of the W most-recent tokens and must offload everything older into its weights — *context becomes weight*. We study three variants: a plain **windowed** model, a **startup** model with a trainable cold-start prompt, and a **persistent-prompt** model whose prompts are always attendable (trainable attention sinks). On Qwen2.5-0.5B, windowed finetuning recovers near-full-context language-modelling quality at a 256-token window and drives the position-1 sink from 57% of heads to 0%; the sink relocates onto the startup prompt. On the hybrid Qwen3.5-9B (only 1/4 of layers are softmax attention), finetuning just those 8 layers under a window matches the model’s full-context perplexity at a 256-token window and *beats StreamingLLM without keeping any sink tokens*. A train-window $\times$ inference-window study yields a simple deployment rule — *train small, deploy anywhere* — and the persistent-prompt variant turns the sink into 64 trainable always-on registers that absorb a controllable 19% of attention.

1 Introduction

Transformer language models are trained with a causal (triangular) attention mask: a query at position t may attend only to keys at positions $\leq t$. A robust empirical phenomenon emerges under this mask — the *attention sink* (Xiao et al. 2024): across heads and layers, a disproportionate share of attention mass lands on the very first token(s) of the sequence, even when those tokens carry no relevant information. The sink is now understood as a “no-op” that heads use to dump attention they do not need, and exploiting it (keeping a few initial tokens) is what lets streaming inference work at all (Xiao et al. 2024; Han et al. 2024).

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We take the sink seriously as a *cause*, not just a curiosity. Two observations motivate this paper. First, the sink is an artifact of the mask geometry: under a triangular mask the first token is in the receptive field of *every* later query, so it is structurally positioned to accumulate attention; the data, by contrast, dilute the first token’s importance as context grows. Second, the sink is exactly what breaks a model when we try to bound its memory. If we want a *constant working memory* — attention restricted to the W most-recent tokens, so that compute and KV-cache are $O(W)$ rather than $O(t)$ — then a plain sliding window deletes position 0, removes the sink, and the pretrained model collapses.

The standard response is to keep the sink at inference time (StreamingLLM: retain a few initial “sink” tokens plus the recent window) (Xiao et al. 2024). We ask a different question: *can we finetune the sink away?* If the model is trained under the sliding-window mask, every token must learn to predict from a bounded window, and any long-range information it still needs has to be written into the *weights* rather than read from far-away keys. This is the sense in which “context is the new weight”: adaptation that an in-context model performs by attending to distant tokens is, after windowed finetuning, performed by the parameters.

We make the following contributions:

- We **disentangle two types of attention sink** that prior work conflates — a *positional* sink on the first token and a *no-op* sink on low-information tokens (commas, periods) — and show they are caused by two *different mechanisms*: the positional sink by the triangular mask’s exposure imbalance, the no-op sink by content-driven bounded-memory token usage (Sec. 2–3). We show by finetuning that the positional sink is *learned and removable* — a windowed mask drives it to zero — while the no-op sink remains and is leaned on more.
- We introduce three windowed-finetuning variants — **windowed**, **startup** (a trainable cold-start prompt), and **persistent-prompt** (always-on trainable registers) — that give a model a constant working memory (Sec. 4).
- On Qwen2.5-0.5B and the hybrid Qwen3.5-9B we show that windowed finetuning recovers near-full-context perplexity at a 256-token window, beats StreamingLLM *without* sink tokens, and obeys a clean “train small, deploy anywhere” rule (Sec. 5).

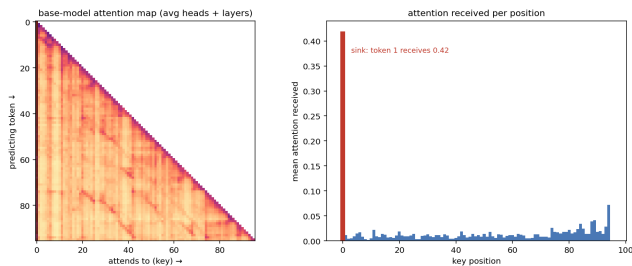


Figure 1: Type 1 (positional) sink in a base causal LM. *Left*: attention map (averaged over heads and layers); the first column is a bright vertical stripe. *Right*: mean attention received per key position — token 1 absorbs 0.42, every other token < 0.05 .

2 Two Types of Attention Sink

We distinguish two distinct phenomena that are both commonly called “attention sinks.” They differ in *where* the attention lands and in *what* drives it, and the rest of the paper turns on telling them apart.

Type 1 — the positional sink. A large, almost content-independent share of every query’s attention falls on the first one or two tokens of the sequence, regardless of their content. Figure 1 shows this for a base model: the leftmost column of the attention map is a bright vertical stripe, and token 1 absorbs on average 0.42 of the attention mass — an order of magnitude more than any later key — across a majority of heads. This is the sink exploited by StreamingLLM (Xiao et al. 2024): it is tied to absolute *position*, not to meaning.

Type 2 — the no-op (low-information) sink. A second sink falls not on the first position but on semantically near-empty tokens *within* the sequence — punctuation such as commas and periods, delimiters, and other filler. These tokens carry little predictive information of their own; a head with nothing to retrieve at a given step dumps its attention onto the nearest such token, using it as a local “no-op” or scratch register. Unlike Type 1, the no-op sink is distributed throughout the text and keyed to token *identity* rather than to absolute position.

Prior work does not disentangle the two. Existing accounts of the attention sink have observed *both* behaviours — the large share of attention parked on the initial token(s) (Xiao et al. 2024; Han et al. 2024) and the attention absorbed by low-information delimiter tokens such as commas and periods — and have gone as far as describing both as “no-op” functionalities a head uses to discard attention. They stop there: the two are folded into a single “attention sink,” neither separated from one another nor traced to distinct causes. Our central claim is that they are in fact *disentangleable* and arise from *two different mechanisms*. Type 1 is an artifact of the triangular mask’s positional exposure imbalance (H1): it is a property of the *architecture*, and it disappears when the mask changes. Type 2 is a content-driven no-op keyed to token *identity* (H2): it is a property of the *data and the model’s use of it*, and it persists — indeed is leaned on *more*

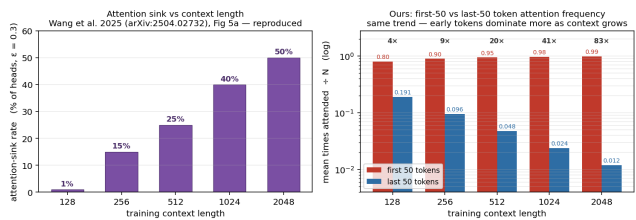


Figure 2: The measured Type 1 sink grows with context length, the opposite of the data-side first-token importance ($\propto 1/C$) — the model sinks *against* the data, an artifact of the mask (H1).

— once Type 1 is removed (H3). Separating the two is what lets us remove the harmful, mask-induced sink while keeping (and exploiting) the useful one. We state these mechanisms as testable hypotheses next.

3 Hypotheses

H1: the positional sink is an artifact of the triangular mask. Under a length- C causal mask a key at position p is attended by the $C - p$ queries after it, so the first token sits in the receptive field of *every* later query — the maximal possible exposure — while its data-side importance is only $\propto 1/C$ and *shrinks* with context. We hypothesize that this exposure imbalance biases training to recruit the earliest positions as a stable, content-free information store, producing the Type 1 sink (a degenerate no-op anchor). Consistent with this, the measured sink *grows* with context (Fig. 2): the model sinks *against* the data-side dilution rather than reflecting it, the signature of an active mechanism rather than a property of the data.

H2: the no-op sink lands on non-meaningful tokens. We hypothesize that Type 2 is the model’s mechanism for “attend to nothing in particular”: heads route attention to low-information tokens — commas, periods, delimiters — as content-based no-op sinks, a behaviour driven by token identity and independent of absolute position.

H3: removing Type 1 makes the model reuse the no-op sinks for storage. If the Type 1 positional bias is removed — e.g. by a windowed objective that masks position 0 out of every deep query’s receptive field — the storage/no-op function is not lost. We hypothesize the model instead repurposes the Type 2 low-information tokens to additionally carry the role the position-0 sink used to play, storing and relaying information on them rather than on a single distant anchor.

H4: no-op sinks are better information relays for streaming. Because Type 2 tokens are distributed through the text — each one sitting at the end of a span of already-accumulated context — information parked on them has a better context-accumulation property than a single far-away position-0 anchor. In a streaming / bounded-window deployment a recent low-information token is always inside the window and has already integrated its local context, making it a better information-relay point than a position-0 sink that the window has scrolled past. The constant-working-memory

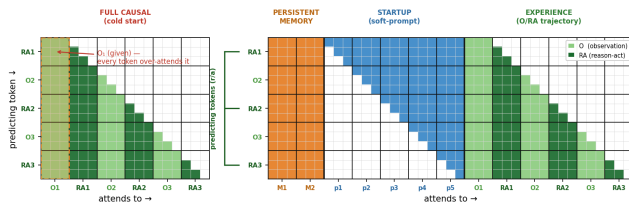


Figure 3: Windowed attention finetuning. The triangular mask (left) is replaced by a sliding window (a band of width W), giving each token a constant working memory of the recent W tokens; older context must be carried by the weights.

regime should therefore not merely tolerate the loss of the Type 1 sink but *benefit* from leaning on Type 2 relays instead.

Why this matters for bounded memory. A constant working memory restricts attention to a window of the W most-recent tokens. For a query deep in the sequence this window no longer contains position 0, so the Type 1 sink is gone; a pretrained model that relies on it produces degenerate distributions and its perplexity explodes. StreamingLLM (Xiao et al. 2024) patches this at inference by always retaining a few initial tokens. We instead remove the dependency by training (Section 4), and ask whether the model falls back on the Type 2 no-op sinks (H3) and whether doing so improves streaming behaviour (H4).

4 Method: Windowed Attention Finetuning

Let the causal mask be $M_{\text{causal}}(t, k) = 1$ iff $k \leq t$. We replace it during finetuning with a sliding-window mask

$$M_{\text{win}}(t, k) = \mathbb{I}[t - W < k \leq t],$$

so each token attends only to the W most-recent positions. We finetune (continued pretraining on generic text, language-modelling loss only — no task supervision) so that the comparison isolates the effect of the *mask*, not of any downstream data. We study three schemes (Fig. 3):

Windowed. The plain sliding window above. The window’s cold start (the first $W-1$ positions, whose window is not yet full) is filled with real preceding tokens during training.

Startup. We prepend P trainable soft prompts that fill the window’s cold start in place of real history, then window over the prompt+content sequence. The prompt slides out of the window for deep tokens, so it functions purely as a learned cold-start anchor — a trainable replacement for the sink token. This grants *availability*: the model can be deployed from token 1 without needing W real tokens of history.

Persistent prompt. We prepend P trainable prompts that are *always* attendable by every content token (a persistent learned prefix), while the content is windowed to W . Each token then sees W recent tokens *plus* P constant learned registers (effective memory $W+P$, of which P is fixed). The prompts become trainable, always-on attention sinks.

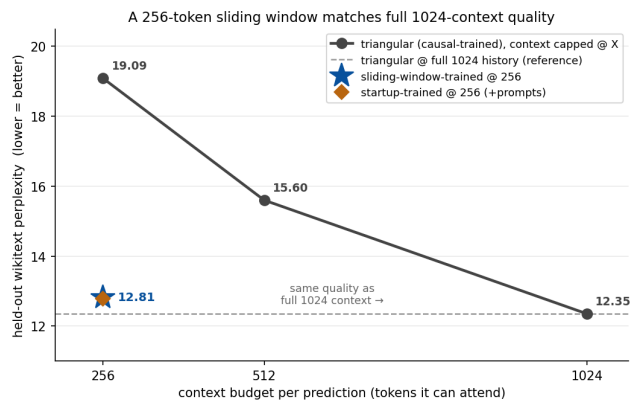


Figure 4: Held-out perplexity vs. deployed working-memory window (Qwen2.5-0.5B). The base model collapses as the window shrinks; the windowed-finetuned model is nearly flat and matches full-context quality at a small window.

model (Qwen2.5-0.5B)	context budget	held-out ppl
triangular (causal-trained)	256	19.09
triangular (causal-trained)	512	15.60
triangular (causal-trained)	1024 (full)	12.35
sliding-window-trained	256	12.81
startup-trained	256 (+ prompts)	12.78

Table 1: Held-out wikttext perplexity vs. context budget on Qwen2.5-0.5B. The causal-trained model needs the full 1024-token history (12.35) and degrades steeply when windowed (19.09 at 256 — worse than the untrained base, because windowing masks the first-token sink it leans on); the windowed-finetuned model reaches 12.81 at a 256-token window, matching the causal model’s full-context quality on a quarter of the context.

5 Experiments

We evaluate on Qwen2.5-0.5B (a pure-softmax-attention model, our proof of concept) and Qwen3.5-9B (the target model). Finetuning is continued pretraining on WikiText-103 (Merity et al. 2017); evaluation is held-out perplexity scored within a working-memory window, plus downstream long-context QA on LongBench (Bai et al. 2024). Unless noted, the window for the headline runs is $W=256$.

5.1 Windowed Finetuning Recovers Full-Context Quality

On Qwen2.5-0.5B, scoring held-out perplexity as a function of the deployed window shows the expected collapse for a base model windowed at inference, and a near-flat curve for the windowed-finetuned model (Fig. 4): a model trained for a 256-token window matches the base model’s full-context quality while attending to $8\times$ fewer tokens (Table 1). Continued pretraining under *full* attention does not help — it amplifies sink reliance — whereas the windowed objective removes it.

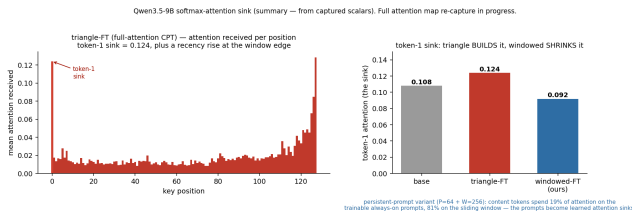


Figure 5: Qwen3.5-9B softmax-layer sink. Full-attention (triangle) finetuning builds the token-1 sink to 0.124; windowed finetuning shrinks it to 0.092; base is between (0.108). The persistent-prompt variant relocates $\sim 19\%$ of every content token’s attention onto trainable always-on registers.

	full attention (learned tendency)	windowed mask (as deployed)
base + history	57.6%	0.0%
sliding-history (trained)	5.8%	0.0%

Table 2: Position-1 sink on Qwen2.5-0.5B given the *same* real history (63 history + 256 content, $W=64$): % of softmax heads with mean attention > 0.3 to the history’s first token. Windowed finetuning cuts the learned sink $\sim 10\times$ ($57.6\% \rightarrow 5.8\%$); the window masks it to 0% at deployment.

5.2 The Sink Is Learned and Removable

Measuring the position-1 sink before and after finetuning shows it is a learned behaviour. On Qwen2.5-0.5B, a majority of softmax heads ($\approx 57\%$) sink onto token 1 in the base and full-attention-finetuned models; windowed finetuning drives this to 0% (Table 2), and the **startup** variant relocates the sink onto the trainable prompt rather than the content. On the larger hybrid model the absolute sink is weaker (see below) but the *direction* is identical: full-attention finetuning *builds* the sink, windowed finetuning *shrinks* it (Fig. 5).

5.3 Where the Attention Goes: Type-1 to Type-2

Removing the Type-1 sink does not destroy the attention mass — it *relocates* it. On Qwen2.5-0.5B we measure, under each scheme’s deployment, the fraction of attention received by the first token (Type-1) versus low-information punctuation tokens (Type-2), and the entropy of the per-position attention (Table 3, Fig. 6). Continued full-causal (triangle) training keeps the Type-1 sink (0.41); windowed and startup training drain it (to 0.15 and 0.008) and the freed mass lands on the distributed Type-2 punctuation tokens ($0.17 \rightarrow 0.37$), with the broadcasting entropy rising $3.9 \rightarrow 5.0$. This is exactly H2–H3: with the positional sink suppressed, the model reuses the content-based no-op tokens and spreads attention more broadly. We note a limit (below): this broader spread is a *distributional* property, not long-range retrieval.

5.4 Scaling to a Hybrid Model

Qwen3.5-9B is a hybrid: of its 32 layers, only 8 (every 4th) are softmax attention; the rest are linear/Mamba-style attention (Gu and Dao 2024; Yang, Kautz, and Hatamizadeh

scheme	Type-1 (pos-0)	Type-2 (punct.)	entropy
base	0.394	0.174	3.94
triangle-CPT	0.412	0.165	3.85
windowed	0.145	0.371	4.60
startup	0.008	0.346	5.00

Table 3: Where attention goes (Qwen2.5-0.5B): fraction received by the position-0 token (Type-1 sink) vs. low-information punctuation tokens (Type-2 no-op sink), and the broadcasting entropy of the per-position attention. Continued full-causal training (triangle) *keeps* the Type-1 sink; windowed/startup training drains it onto the distributed Type-2 tokens and raises the entropy — the spread predicted by H2–H3.

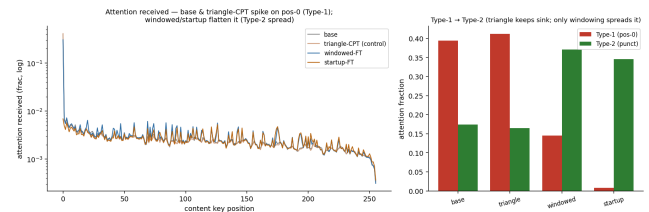


Figure 6: Windowed/startup training drains the Type-1 positional sink (left, log scale: base spikes ≈ 0.39 at position 0; windowed/startup flatten it) and redistributes the mass onto distributed Type-2 punctuation tokens (right), broadcasting entropy rising $3.9 \rightarrow 5.0$.

2024). The sink, and the windowing question, therefore concern only those 8 layers — $3/4$ of the network is already a constant-memory mechanism. We full-finetune just the 8 softmax blocks (freezing the 24 linear layers; 8-bit AdamW with gradient checkpointing on a single 48 GB GPU). Even on this model, bounding the softmax window to 256 tokens collapses the base from a full-context perplexity of 6.5 to 20.0; windowed finetuning restores it to 6.3 — full-context quality at a 256-token softmax window — and, unlike the baselines, gains nothing from StreamingLLM sink tokens because it no longer depends on them (Table 4).

5.5 Train Small, Deploy Anywhere

Evaluating every windowed model at every inference window (Table 4) reveals an asymmetry. A model trained at a *small* window (windowed-128) is nearly flat across *all* deployment windows ($7.44 \rightarrow 6.62$): having learned to predict from 128 tokens, it is robust to any budget. A model trained at a *wide* window is fragile when deployed narrower (windowed-1024 rises to 24.7 at deploy-128), because it learned to lean on context that is no longer present. A model is always fine deployed *wider* than it was trained, and breaks deployed *narrower*: the practical rule is to train at the smallest window one might deploy.

5.6 A Strict Softmax-vs-Gated Control

Is the gain from *windowing*, or merely from more finetuning? We answer with a strict control: two 1.3B models from the

model	perplexity at inference window				
	128	256	512	1024	2048
<i>untrained baseline</i>					
base 9B	59.8	28.1	13.8	7.80	7.26
+ StreamingLLM	9.33	8.47	7.91	7.58	7.28
<i>continued-pretraining baseline</i>					
triangle-FT	33.7	18.2	8.66	6.96	6.59
+ StreamingLLM	7.88	7.39	7.03	6.83	6.56
<i>ours — windowed (one model per training window)</i>					
windowed-128	7.44	7.14	6.94	6.83	6.62
windowed-256	7.82	7.19	6.93	6.79	6.57
windowed-512	11.5	7.82	6.95	6.79	6.55
windowed-1024	24.7	12.7	7.44	6.83	6.54
windowed-2048	27.6	14.5	7.76	6.84	6.53
<i>ours — startup (windowed + cold-start prompt)</i>					
startup-256	7.82	7.17	6.92	6.79	6.57

Table 4: Qwen3.5-9B held-out perplexity, train-window (rows) $\times$ inference-window (columns); bold is native (train = deploy). Windowed models beat both baselines and their StreamingLLM versions at tight windows, with no sink tokens. Small-window models are robust to any deployment window; wide-window models break when deployed narrower (*train small, deploy anywhere*).

flash-linear-attention suite trained on the *same* 100B tokens with the *same* skeleton, differing only in the token-mixer — a softmax Transformer and a gated-linear (GLA) model. Under a 256-token sliding window the softmax base collapses ($9.94 \rightarrow 16.13$); continued *full-causal* pretraining does not fix it (13.53, the control); windowed finetuning recovers it to 10.16, near its own full-context 9.12 and approaching the natively constant-memory gated model’s 8.93 (Table 5). So windowed *training* turns a softmax model into something close to a gated/constant-memory model — the recovery is from the window, not the extra training.

Compute. The payoff is efficiency. Full softmax attention is $O(L^2)$; the windowed and gated mechanisms are $O(L)$. At a 32k context the windowed model runs $2.7\times$ faster than full attention (Table 6), with linear rather than quadratic growth.

5.7 Persistent Prompts as Trainable Sinks

The persistent-prompt variant ($P=64$ always-on prompts + a $W=256$ window) lets content tokens spend a controllable fraction of attention on the learned registers: in our run, 19% of every content token’s attention goes to the 64 persistent prompts and 81% to the window. The prompts thus become explicit, trainable attention sinks — the function the first token was being abused for, now a designed component. On perplexity the persistent and startup variants track the plain windowed model, with a small edge when deployed *narrower* than trained (the prompt fills the coldest part of the window). The registers matter most on a downstream task that needs distant context. On multi-hop QA (HotpotQA, Qwen2.5-0.5B) under a 256-token window, a purely windowed model loses far context (answer-perplexity 38.2; the untrained base it rescues is at 120.4, and that base patched with StreamingLLM at 15.8), whereas the persistent-prompt

mixer	finetune	windowed@256	full
softmax	base	16.13	9.94
	triangle-CPT	13.53	8.93
	windowed	10.16	9.12
	startup	10.19	9.15
gated (GLA)	base	—	10.41
	triangle-CPT	8.93	8.93

Table 5: Strict same-skeleton, same-(100B-token) control (flash-linear-attention 1.3B): a softmax Transformer vs. a gated-linear (GLA) model where *only the token-mixer differs*. Held-out perplexity under a 256-token sliding window. The softmax base collapses ($9.94 \rightarrow 16.13$); full-attention continued pretraining (triangle-CPT) does *not* fix it (13.53); **windowed finetuning recovers it** (10.16, near its own full-context 9.12), approaching the natively constant-memory gated model (8.93). Recovery is window-robust: windowed-FT at $W=128/256/512$ gives 11.2/10.2/9.5 vs. base 18.9/16.1/14.1.

seq. length	full	windowed	gated
8k	313	226	258
16k	797	436	499
32k	2363	866	989

Table 6: Forward latency (ms) vs. sequence length (fla 1.3B, one A6000). Full softmax attention scales quadratically; windowed and gated scale linearly — at 32k the windowed model is $2.7\times$ faster than full.

model recovers much of it (23.9) — behaving as a *trained* StreamingLLM that keeps a global anchor as the window slides. This also marks the boundary of the broadcasting effect of §5: windowed training spreads attention onto the Type-2 tokens (a distributional change) but does not by itself relay a *specific* distant fact; the persistent registers are what restore long-range anchoring, unifying constant-memory streaming with retrieval in one trained model.

5.8 Downstream Long-Context QA

To isolate the mask effect from task learning we also continued-pretrain (no QA supervision) and evaluate zero-shot on LongBench multi-hop QA (Bai et al. 2024). At a tight 256-token window, the windowed model uses its bounded memory markedly better than a full-attention-pretrained control, and StreamingLLM helps the baselines but not the windowed model — the same signature as in perplexity, on a real task (Fig. 7).

6 Related Work

Attention sinks and streaming. Xiao et al. (2024) identified the sink and exploited it for streaming inference by retaining initial tokens; Han et al. (2024) use a similar fixed pattern for length generalization. These keep the sink at inference; we remove the dependency by finetuning, and (in the persistent-prompt variant) replace the accidental sink with

The NMR test — continued pretraining on a windowed QA dataset (all other settings, only the attention mask differs), then zero-shot LongBench multi-hop QA at a fixed working memory. At a tight 256-token window, windowed CPT > Full CPT — it decreases the triangle-mean loss with no task training.

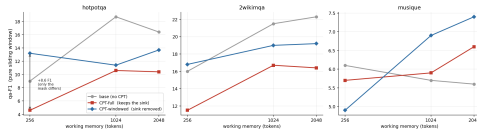


Figure 7: LongBench multi-hop QA (Qwen2.5-0.5B, continued-pretraining, no task supervision). At a tight window the windowed model uses a bounded working memory better than a full-attention control; the gap shrinks as the window grows.

trainable registers.

Windowed and efficient attention. Sliding-window attention is used architecturally by Longformer (Beltagy, Peters, and Cohan 2020) and Mistral (Jiang et al. 2023), and linear/state-space models (Gu and Dao 2024; Yang, Kautz, and Hatamizadeh 2024) achieve constant memory by construction; Qwen3.5 is a hybrid of the two. We instead *finetune* a pretrained softmax model into a windowed one and study what the sink does under that change. Position-extrapolation methods such as ALiBi (Press, Smith, and Lewis 2022) share the “train short, test long” spirit; we observe a complementary “train small, deploy anywhere” rule for the attention window.

Context as weight. Our framing — that windowing forces older context into the parameters — connects to context distillation and knowledge editing; here the mechanism is attention geometry rather than an explicit distillation loss.

7 Conclusion

The attention sink is a learned consequence of the triangular causal mask, and it is exactly what makes a pretrained model fragile to a bounded context. Finetuning under a sliding-window mask removes the dependency: the model acquires a constant working memory, matches full-context quality at a small window, and — unlike inference-time fixes — needs no sink tokens. Two design points fall out: “train small, deploy anywhere,” and the persistent-prompt variant that turns the accidental sink into a controllable set of trainable registers. We view this as evidence that, for the long-range part of a model’s behaviour, context can be traded for weight.

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